



A comprehensive experimental and kinetic modeling study of tert-butanol combustion



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ABSTRACT

The combustion of tert-butanol was studied experimentally in this work, including flow reactor pyrolysis at 30–760 Torr, laminar flat premixed flame at 30 Torr, coflow methane/tert-butanol diffusion flame at atmospheric pressure, and laminar flame speed at 1–10 atm. Synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) was applied in the investigation of pyrolysis and flame, and the schlieren technique was used in the measurement of laminar flame speed. A detailed kinetic model consisting of 209 species and 1486 reactions was developed in this work, which can predict the decomposition of tert-butanol and the formation of aromatics over wide temperature and pressure ranges. Kinetic analysis illustrates that the unimolecular decomposition of tert-butanol is the most important and sensitive reaction class in different combustion modes, which readily produce the most important aromatic precursors: tert-butanol → iso-butene → 2-methylallyl → allene → propargyl. As a result, the production of benzene, phenyl radical, and benzyl radical are all enhanced in the combustion of tert-butanol compared to those of other butanol isomers. The self-combination of propargyl and the combination of propargyl with other resonantly stabilized radicals are the key steps of aromatics formation in the combustion of tert-butanol.

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1. Introduction

Biofuels such as alcohols are promising future fuels, which can offset CO₂ and soot emission to reduce greenhouse effect and air pollution, etc. [1]. Butanols have several remarkable advantages over ethanol as alternatives to petroleum-based transportation fuels, such as higher energy density, better miscibility with practical fuels, and lower water affinity. Therefore increasing attention has been paid to the investigations of their combustion properties, including flame propagating speed [2–4], ignition delay time [5–8], pollutants emission [9–12], etc. The butanol isomers behave very differently due to the significant influence of isomeric structures. Among the butanol isomers, tert-butanol (tC₄H₉OH) has the most branched molecular structure. It has a high octane number

(RON = 107 [13]), thus has been used for decades as an octane booster to prevent knock in spark-ignition engines [14,15].

A few studies were performed previously on the combustion chemistry of tert-butanol [3,5,7,16–23]. Yasunaga et al. [7] studied the pyrolysis and oxidation of tert-butanol using shock tubes, in the pressure range of 1.2–2.2 atm and temperature range of 1000–1800 K. Gas chromatography was applied to measure the stable products. Cai et al. [18] studied the pyrolysis of tert-butanol in a flow reactor at low pressure using synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). Both free radicals and stable intermediates were observed. Recently, Lefkowitz et al. [19] also studied the oxidation of tert-butanol in a flow reactor at the pressure of 12.5 atm and in the temperature range of 675–950 K. They found that tert-butanol does not exhibit low temperature oxidation behavior under their experimental conditions, because no negative temperature coefficient region was observed. Welz et al. [24] studied the low-temperature oxidation of tert-butanol at a very low pressure (4–5.1 Torr). The reaction of iso-butyl-2-ol radical (tC₄H₈OH) and O₂ forming OH radical was concluded as the dominant chain-propagating channel in their work. Yang et al. [16] focused on the qualitative identification of

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Table 1
Experimental conditions of laminar flames.

Laminar flame speed								
Name	P^a	T^b	ϕ	$X_{\text{C}_4\text{H}_9\text{OH}}$	X_{O_2}	X_{N_2}	X_{He}	
FS1	1.0	423	0.7–1.5	1.00	0.21	0.79	–	
FS2	2.0	423	0.7–1.5	1.00	0.21	0.79	–	
FS3	5.0	423	0.7–1.3	1.00	0.21	0.79	–	
FS4	10.0	423	0.7–1.0	1.00	0.21	0.79	–	
FS5	10.0	423	0.8–1.4	1.00	0.14	–	0.86	
FS6	20.0	423	0.7–1.4	1.00	0.14	–	0.86	
Burner stabilized premixed flames ^c								
Name	P^a	$\dot{m}^d$	ϕ	X_{Ar}	X_{O_2}	$X_{\text{C}_4\text{H}_9\text{OH}}$		
PF1	0.0395	0.003187	1.30	0.5000	0.4108	0.0892		
PF2	0.0395	0.003520	1.80	0.5000	0.3846	0.1154		
Coflow diffusion flames ^e								
Name	P^a	Q_{air}^f	Q_{Ar}^f	$Q_{\text{N}_2}^f$	$Q_{\text{CH}_4}^f$	$Q_{\text{C}_4\text{H}_9\text{OH}}^f$	v_{air}^g	v_{fuel}^g
CF1	1.0	80,000	5.87	438	160	24.50	16.48	13.33
CF2	1.0	80,000	5.87	401	209	12.25	16.48	13.33

^a P , initial pressure of premixed gases for flame speed measurements, environment pressure for burner stabilize flames and coflow diffusion flames, unit: atm.

^b T , initial temperature of premixed gases, unit: K.

^c PF: abbreviation for burner stabilized premixed flames;

^d $\dot{m}$, mass flow rate, unit: g/(s cm²).

^e CF: abbreviation for coflow diffusion flame.

^f Q_i , flow rate of species i , unit: SCCM; the gas temperature for fuel mixture and air are 493 and 298 K, separately.

^g v_i , converted into mean velocity of fuel/oxidizer stream at 273 K, unit: cm/s.

intermediates in the rich flames of four butanol isomers ($\phi = 1.7$). Oßwald et al. [17] then measured the mole fractions of major flame species and intermediates in the flames of four butanol isomers at the same equivalence ratio of 1.7. Furthermore, Moss et al. [5], Yasunaga et al. [7] and Stranic et al. [20] measured the ignition delay times of tert-butanol at different temperatures, pressures and equivalence ratios, while Wu et al. [2] and Veloo et al. [3] measured laminar flame speeds of tert-butanol. tert-Butanol, has a much higher sooting tendency than its isomers [25,26]. However, no kinetic model for tert-butanol combustion has been comprehensively validated over wide pressure and temperature ranges. The formation mechanism of the polycyclic aromatic hydrocarbon (PAH) in the combustion of tert-butanol remains unclear. Therefore, refinements on current tert-butanol models are urgently needed to expand their capability to predict the combustion of tert-butanol and the formation of PAH in this process over wide pressure and temperature ranges.

In this work, the pyrolysis of tert-butanol was studied in a flow reactor in the temperature range of 850–1550 K and at pressures of 30, 150 and 760 Torr. The laminar flame speeds of tert-butanol were measured at pressures of 1–20 atm. These experiments focused on the pressure effect upon the combustion of tert-butanol. Two rich laminar premixed $\text{tC}_4\text{H}_9\text{OH}/\text{O}_2/\text{Ar}$ flames at 30 Torr and two coflow methane/tert-butanol diffusion flames at atmospheric pressure were also studied using SVUV-PIMS. A model consisting of 209 species and 1486 reactions was developed based on our previous butanol and methane models [18,27]. A comprehensive model validation and detailed kinetic analysis on the high temperature combustion of tert-butanol will be presented in the following paragraphs.

2. Experimental methods

Flow reactor pyrolysis, premixed flat flame, coflow diffusion flame, and laminar flame speed were all carried out at National Synchrotron Radiation Laboratory in Hefei, China. All of the experimental conditions are listed in Table 1 (except pyrolysis). The purities of the samples, including CH_4 , N_2 , O_2 , Ar, He and $\text{tC}_4\text{H}_9\text{OH}$, are

99.995%, 99.999%, 99.999%, 99.99%, 99.99% and 99%, respectively. The gas flow rates were regulated using mass flow controllers, and tert-butanol was injected into a vaporizer with a chromatography pump. The SVUV-PIMS system, including the synchrotron beam-line and the experimental apparatuses, were introduced in detail in previous studies [10,28–32]. Here, we will briefly introduce each of them in this section.

The uncertainty in the SVUV-PIMS measurements is related to the sampling process and the photoionization cross sections (PICS) of the combustion species. Generally, it is within $\pm 10\%$ for combustion species quantified via cold gas calibration, $\pm 25\%$ for stable combustion species with known PICSs, and about a factor of 2 for free radicals and combustion species with estimated PICSs [10,33]. The PICSs of combustion species are available online in the database of [34].

2.1. Pyrolysis in a flow reactor

The flow reactor pyrolysis apparatus consists of a pyrolysis chamber with an electrically heated furnace, a differentially pumped chamber with molecular beam sampling system and a photoionization chamber with a home-made reflectron time-of-flight mass spectrometer (RTOF-MS). The flow reactor is equipped in the pyrolysis chamber. It contains a 6.8-mm-ID (inner diameter) α -alumina tube, which is heated by a furnace over a length of 150 mm. α -Alumina is thermally stable, the catalytic effect of which been evaluated to be very weak in pyrolysis experiments [35]. The pyrolysis products are sampled 10 mm downstream of the outlet of the reactor using a quartz conical nozzle. The molecular beam that is formed by the sampled gas is then ionized by the tunable SVUV light, and detected by RTOF-MS.

The pyrolysis of tert-butanol was studied at different pressures (30, 150 and 760 Torr), while the inlet compositions of reactant and diluent gas remained constant (3% $\text{tC}_4\text{H}_9\text{OH}$ and 97% Ar). tert-Butanol was vaporized at 393 K, mixed with the carrier gas (Ar), then injected to the flow reactor. Temperature distributions along the centerline of the flow reactor were measured with a Pt-6%Rh/Pt-30%Rh thermocouple, which are provided as Supplementary material. The measured temperature profiles were used

as the input parameters in the simulation [30]. The maximum value ($T_{\max}$) of each profile is used as the X-axis for plotting the measurements and predictions. The uncertainty of $T_{\max}$ is within ± 30 K.

2.2. Burner stabilized premixed flame

Rich laminar premixed tert-butanol/ O_2 /Ar flat flames ($\phi = 1.3$ and 1.8) were stabilized at 30 Torr on a 60 mm diameter McKenna burner. The burner can be moved by a step motor with high precision (<0.05 mm). Therefore, species profiles were obtained as a function of the height above the burner (HAB). The sampling and detecting procedures are same as those in the pyrolysis experiment, and will not be repeated here. Intermediates identification and mole fraction evaluation were based on previously reported methods [36,37]. The flame temperature profile was measured using 0.1-mm-diameter Pt-6%Rh/Pt-30%Rh thermocouples coated with Y_2O_3/BeO anti-catalytic ceramic [38,39], which was calibrated for radiative heat loss [40] and the cooling effect of sampling nozzle [41]. The uncertainty of the maximum flame temperature was estimated to be within ± 100 K.

2.3. Co-flow diffusion flame

The co-flow diffusion flame burner consists of a 10-mm-ID stainless steel fuel tube nested in the center of a 102-mm-ID co-flow air tube. It is fixed to a two dimensional translational stage that can be moved with high precision (<0.05 mm) to change the sampling position. Flame species along the centerline of the coflow flame were sampled using a quartz probe, except in the region near the bottom of the flame, due to the shape limit of the sampling probe. The details of the experimental method can be found in a previous study [10]. The inlet mole fractions of the mixtures in the fuel tube, including the fuels (CH_4 and tC_4H_9OH), diluent gas (N_2), and calibration gas (Ar), were varied in two different flame conditions, while the carbon flux was kept identical. The mean velocities of the fuel mixture and air in the standard condition were 13.33 and 16.48 cm/s in both flames.

2.4. Laminar flame speed

The laminar flame speeds of tert-butanol were measured using the schlieren technique in an electrically heated constant-volume cylindrical combustion vessel (150 mm in diameter and 152 mm in length). Premixed combustible mixtures of tC_4H_9OH , O_2 and inert gas (N_2 or He) were prepared via the partial pressure method in a large premixing vessel, in order to keep an identical composition of the combustible mixture in the measurements at different pressures at one specific equivalence ratio. The combustion vessel, the premixing vessel and the connecting pipes were heated to 423 K during the experiments. After the premixed gas was spark-ignited, a spherical flame propagated outward. This process was recorded using a high speed camera (Phantom V310) of the schlieren system, which was built up through a 75-mm-diameter aperture in the combustion vessel. The spatial resolution of the camera is approximately 0.156 mm/pixel, and the recording speed is 10,000 frames/s.

Two groups of experiments were performed in this work. The tert-butanol/air mixtures were used at pressures of 1–10 atm; and the tert-butanol/ O_2 /He mixtures were studied at higher pressures of 10 and 20 atm. The measurement of each flame condition was repeated three times. A non-linear extrapolation method [42,43] was adopted in this work. The detailed data processing procedure can be found in the Supplementary material. The uncertainty of the measured laminar flame speed is ± 0.83 to ± 2.27 cm/s depending on the equivalence ratios and initial pressures, which is presented along with experimental data in the corresponding figures.

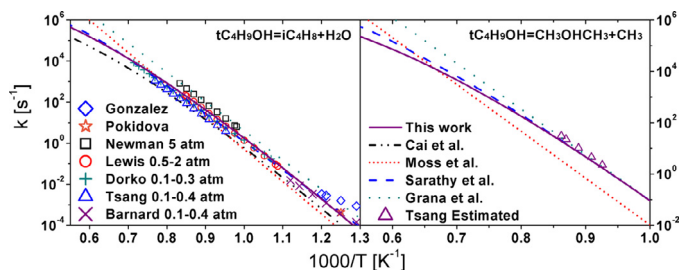


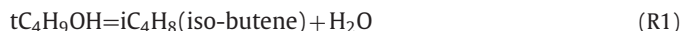
Fig. 1. Plot of the rate constants of (a) (R1) and (b) (R2). The lines are the values used in the present model (solid line) and previous models [5,18,47,48] at 1 atm. The symbols are the experimental data from literatures [23,66–69].

3. Kinetic model construction

The present model was mainly developed from our previously reported low-pressure combustion model of tert-butanol [18] and recently reported methane model [27]. Although the PAH sub-mechanism was introduced in detail in our previous studies [27], it will be described again here for the convenience of the sequent discussions. Important reaction classes among the decomposition process of tert-butanol are listed in Table 2. The thermodynamic and transport data of most species were taken from [18,27,29,44,45] or estimated following the procedure described in [46]. The reaction mechanism, thermodynamic data, and transport data files in CHEMKIN format are available as Supplementary material.

3.1. Sub-mechanism of tert-butanol

Two possible classes of unimolecular decomposition reactions are included in the sub-mechanism of tert-butanol, including water elimination (R1) and bond dissociation ((R2)–(R4)). According to the ab initio calculation and rate constant evaluation via Rice–Ramsperger–Kassel–Marcus (RRKM) theory in the work of Cai et al. [18], (R1) and (R2), with relatively low energy barriers, provide a dominant contribution in the decomposition of tert-butanol. As shown in Fig. 1, the rate constant of (R2) in the present model was adopted from the calculation of Cai et al. [18], while that for (R1) was re-evaluated from previous models [5,47,48] and experimental measurements [23,66–69].



H-atom abstraction of tert-butanol by a series of free radicals was considered in the present model, and most of their rate constants were taken from the model of Sarathy et al. [47]. The rate constants of H-atom abstraction reactions by OH radical were re-fitted in this work. Figure 2 presents the total rate constant of this reaction, summarizing the two channels producing iso-butyl-2-ol (tC_4H_8OH) and tert-butyl-hydroxy (tC_4H_9O) radicals ((R5) and (R6)). Stranic et al. [70] measured the rate constant of this reaction in the temperature region of 900–1200 K, which provided the only set of data available under this condition. However, the value recommended in their work showed large discrepancy with former experimental results at lower temperature [71–73]. In the present model, a new rate constant was evaluated from the experimental data of Stranic et al. [70] and the former studies [71–73], as shown in Fig. 2. The branching ratio between R5 and R6 was taken

Table 2Selected reactions in the present model^a.

	Reactions		A	n	E _a	Pressure [Torr]	Ref.
R1	tC ₄ H ₉ OH=iC ₄ H ₈ +H ₂ O						[18]
		PLOG	1.18 × 10 ⁹³	−23.24	106,272	5	
		PLOG	3.51 × 10 ⁸⁶	−21.17	104,468	30	
		PLOG	2.19 × 10 ⁸²	−19.88	102,766	70	
		PLOG	2.23 × 10 ⁷⁶	−18.08	100,065	200	
		PLOG	3.84 × 10 ⁶⁷	−15.47	95,722	760	
R2	tC ₄ H ₉ OH=CH ₃ COHCH ₃ +CH ₃						[18]
		PLOG	5.93 × 10 ⁵¹	−10.80	87,322	7600	
		PLOG	2.80 × 10 ³⁸	−6.80	81,200	76,000	
		PLOG	7.75 × 10 ¹¹⁵	−29.60	129,549	5	
		PLOG	1.40 × 10 ¹¹⁴	−28.74	132,060	30	
		PLOG	9.09 × 10 ¹¹¹	−27.96	132,301	70	
R3	tC ₄ H ₉ OH(+M)=tC ₄ H ₉ +OH(+M)						[18]
		PLOG	4.98 × 10 ¹⁰⁷	−26.56	131,496	200	
		PLOG	1.10 × 10 ¹⁰⁰	−24.19	128,841	760	
		PLOG	7.33 × 10 ⁹⁰	−21.20	127,322	7600	
		PLOG	5.10 × 10 ⁷⁸	−17.40	123,300	76,000	
		PLOG	1.20 × 10 ²⁴	−2.13	97,281		
R4	tC ₄ H ₉ OH(+M)=H+tC ₄ H ₈ OH(+M)						[47]
		LOW /	4.53 × 10 ⁷⁴	−16.28	107,810	/	
		TROE /	0.37, 469.09, 2.9 × 10 ¹² , 3672.8			/	
		LOW /	1.30 × 10 ¹⁸	−0.45	104,460		
		TROE /	1.54 × 10 ⁶⁵	−13.67	114,110	/	
		TROE /	0.32, 507.26, 8 × 10 ⁶ , 2535.1			/	
R5	tC ₄ H ₉ OH+OH=H ₂ O+tC ₄ H ₈ OH		2.88 × 10 ⁴	2.70	−800		b
R6	tC ₄ H ₉ OH+OH=H ₂ O+tC ₄ H ₉ O		1.20 × 10 ³	2.70	−800		b
R7	tC ₄ H ₉ OH+H=H ₂ +tC ₄ H ₈ OH		2.17 × 10 ⁷	2.00	6526		[48]
R8	tC ₄ H ₉ OH+H=H ₂ +tC ₄ H ₉ O		2.41 × 10 ⁶	2.00	6526		[48]
R9	tC ₄ H ₉ OH+CH ₃ =CH ₄ +tC ₄ H ₈ OH		1.36	3.65	9150		[47]
R10	tC ₄ H ₉ OH+CH ₃ =CH ₄ +tC ₄ H ₉ O		1.02	3.57	8220		[47]
R11	tC ₄ H ₈ OH=iC ₄ H ₈ +OH						[18]
		PLOG	1.94 × 10 ²⁴	−4.97	18,891	5	[18]
		PLOG	2.14 × 10 ²⁸	−5.88	22,103	30	
		PLOG	8.63 × 10 ³⁰	−6.49	24,265	80	
		PLOG	5.58 × 10 ³²	−6.92	25,805	150	
		PLOG	6.14 × 10 ³⁷	−8.12	30,275	760	
		PLOG	2.71 × 10 ⁴⁶	−10.25	38,119	7600	
R12	tC ₄ H ₈ OH=iC ₃ H ₅ OH+CH ₃						[18]
		PLOG	1.24 × 10 ⁴⁸	−10.39	42,159	76,000	
		PLOG	9.85 × 10 ²³	−5.32	20,772	5	
		PLOG	6.58 × 10 ²⁸	−6.36	24,324	30	
		PLOG	7.72 × 10 ³¹	−7.05	26,758	80	
		PLOG	1.06 × 10 ³⁴	−7.54	28,514	150	
R13	CH ₃ COCH ₃ +CH ₃ =tC ₄ H ₉ O						[47]
		PLOG	1.19 × 10 ⁴⁰	−8.96	33,731	760	
		PLOG	8.28 × 10 ⁵⁰	−10.62	43,345	7600	
		PLOG	7.26 × 10 ⁵²	−10.75	48,293	76,000	
		PLOG	1.50 × 10 ¹¹	0.00	11,900		
		PLOG	1.37 × 10 ⁹	1.46	1355		
R14	CH ₃ COCH ₃ +H(+M)=CH ₃ COHCH ₃ (+M)						[47]
		LOW /	2.03 × 10 ³⁹	−6.64	5769	/	
		TROE /	−0.569, 299.0, 9147.0, −152.4			/	
		H ₂ /2.0/ H ₂ O/6.0/ CH ₄ /2.0/ CO/1.5/ CO ₂ /2.0/ C ₂ H ₆ /3.0/ AR/0.7/					
		LOW /	1.37 × 10 ⁹	1.46	1355		
		TROE /	2.03 × 10 ³⁹	−6.64	5769	/	
R15	iC ₃ H ₅ OH+H(+M)=CH ₃ COHCH ₃ (+M)						[49]
		LOW /	2.03 × 10 ³⁹	−6.64	5769	/	
		TROE /	−0.569, 299.0, 9147.0, −152.4			/	
		H ₂ /2.0/ H ₂ O/6.0/ CH ₄ /2.0/ CO/1.5/ CO ₂ /2.0/ C ₂ H ₆ /3.0/ AR/0.7/					
		LOW /	1.37 × 10 ⁹	1.46	1355		
		TROE /	2.03 × 10 ³⁹	−6.64	5769	/	
R16	iC ₃ H ₅ OH=CH ₃ COCH ₃						[49]
		PLOG	7.42 × 10 ⁴⁶	−10.56	67,420	76	
		PLOG	4.42 × 10 ⁴²	−9.09	67,069	760	
		PLOG	2.90 × 10 ²⁷	−4.35	61,613	76,000	
		PLOG	7.42 × 10 ⁴⁶	−10.56	67,420	76	
		PLOG	4.42 × 10 ⁴²	−9.09	67,069	760	
R17	iC ₃ H ₅ OH+H(+M)=CH ₃ COCH ₃ +H(+M)						[7]
		PLOG	2.90 × 10 ²⁷	−4.35	61,613	76,000	
		TCHEB /	270	3000	/		
		PCHEB /	9.87 × 10 ^{−3}	9.87 × 10 ²	/		
		CHEB /	5	3	/		
		CHEB /	11.686	−0.893	−0.190	/	
R18	iC ₄ H ₈ +H(+M)=iC ₄ H ₉ (+M)						[7]
		CHEB /	1.026	0.881	0.113	/	
		CHEB /	−0.108	0.154	0.106	/	
		CHEB /	−0.115	−0.669	0.014	/	
		CHEB /	−0.048	−0.065	−0.025	/	
		CHEB /	1.61 × 10 ¹³	0.00	4398		
R19	tC ₄ H ₉ (+M)=iC ₄ H ₈ +H(+M)						[7]
		LOW /	6.26 × 10 ³⁸	−6.66	7000	/	
		TROE /	1.0, 1000, 1310, 48,097			/	
		H ₂ /2.0/ H ₂ O/6.0/ CH ₄ /2.0/ CO/1.5/ CO ₂ /2.0/ C ₂ H ₆ /3.0/ AR/0.7/					
		LOW /	5.66 × 10 ¹¹	0.70	36,570		
		TROE /	1.90 × 10 ⁴¹	−7.36	36,632	/	
R20	H ₂ /2.0/ H ₂ O/6.0/ CH ₄ /2.0/ CO/1.5/ CO ₂ /2.0/ C ₂ H ₆ /3.0/ AR/0.7/						[7]
		LOW /	0.293, 649, 60,000, 3425.9			/	
		TROE /				/	
		H ₂ /2.0/ H ₂ O/6.0/ CH ₄ /2.0/ CO/1.5/ CO ₂ /2.0/ C ₂ H ₆ /3.0/ AR/0.7/					
		LOW /				/	
		TROE /				/	

(continued on next page)

Table 2 (continued)

	Reactions		A	n	E _a	Pressure [Torr]	Ref.
R20	C ₃ H ₆ + CH ₃ (+M)=iC ₄ H ₉ (+M)		1.89 × 10 ³	2.67	6850		b
		LOW /	1.30 × 10 ²⁸	−4.27	2431.1	/	
		TROE /	0.565, 60,000, 534.2, 3007.2			/	
R21	H ₂ /2.0/ H ₂ O/6.0/ CH ₄ /2.0/ CO/1.5/ CO ₂ /2.0/ C ₂ H ₆ /3.0/ AR/0.7/						
	iC ₄ H ₉ + H=iC ₄ H ₈ + H ₂		9.03 × 10 ¹¹	0.00	0		[50]
R22	iC ₄ H ₉ + H=iC ₃ H ₇ + CH ₃						[50]
		PLOG	1.10 × 10 ³²	−5.04	16,760	76	
		PLOG	1.90 × 10 ³⁵	−5.83	22,470	760	
		PLOG	1.35 × 10 ⁴⁰	−7.02	31,000	7600	
R23	tC ₄ H ₉ + H=iC ₃ H ₇ + CH ₃						[50]
		PLOG	2.80 × 10 ³⁴	−5.69	20,500	76	
		PLOG	2.60 × 10 ³⁶	−6.12	25,640	760	
R24	iC ₄ H ₉ =tC ₄ H ₉		5.00 × 10 ¹²	0.00	38,000		[51]
R25	iC ₄ H ₇ + H(+M)=iC ₄ H ₈ (+M)		2.00 × 10 ¹⁴	0.00	0		[52]
		LOW /	1.33 × 10 ⁶⁰	−12.00	5968	/	
		TROE /	0.02, 1096.6, 1096.6, 6859.5			/	
	H ₂ /2.0/ H ₂ O/6.0/ CH ₄ /2.0/ CO/1.5/ CO ₂ /2.0/ C ₂ H ₆ /3.0/ AR/0.7/						
R26	CH ₃ CCH ₂ + CH ₃ (+M)=iC ₄ H ₈ (+M)		2.50 × 10 ¹³	0.00	0		[52]
		LOW /	1.33 × 10 ⁶⁰	−12.00	5968	/	
		TROE /	0.02, 1096.6, 1096.6, 6859.5			/	
	H ₂ /2.0/ H ₂ O/6.0/ CH ₄ /2.0/ CO/1.5/ CO ₂ /2.0/ C ₂ H ₆ /3.0/ AR/0.7/						
R27	iC ₄ H ₈ + H=iC ₄ H ₇ + H ₂		1.72 × 10 ¹⁴	0.00	8000		[7]
R28	iC ₄ H ₈ + H=iC ₄ H ₇ −1 + H ₂		1.20 × 10 ¹⁴	0.00	13,000		[7]
R29	iC ₄ H ₈ + CH ₃ =iC ₄ H ₇ + CH ₄		4.40	3.50	5675		b
R30	iC ₄ H ₈ + CH ₃ =iC ₄ H ₇ −1 + CH ₄		2.00 × 10 ¹²	0.00	15,000		[7]
R31	iC ₄ H ₈ + OH=iC ₄ H ₇ + H ₂ O		1.50 × 10 ⁸	1.53	775		b
R32	iC ₄ H ₈ + OH=iC ₄ H ₇ −1 + H ₂ O		1.50 × 10 ¹²	0.0	3060		b
R33	iC ₄ H ₈ + H=C ₃ H ₆ + CH ₃						[52]
		PLOG	8.80 × 10 ¹⁶	−1.05	6461	76	
		PLOG	8.00 × 10 ²¹	−2.39	11,180	760	
		PLOG	3.30 × 10 ²⁴	−3.04	15,610	7600	
R34	iC ₄ H ₈ + OH=iC ₃ H ₅ OH + CH ₃		5.00 × 10 ¹¹	0.00	2000		b
R35	iC ₄ H ₈ + O=iC ₄ H ₇ + OH		2.90 × 10 ⁵	2.50	3640		b
R36	iC ₄ H ₈ + O=iC ₃ H ₇ + HCO		3.50 × 10 ⁷	1.65	−972		b
R37	aC ₃ H ₄ + CH ₃ =iC ₄ H ₇		2.00 × 10 ¹¹	0.00	7500		[53]
R38	iC ₄ H ₇ −1=pC ₃ H ₄ + CH ₃		1.20 × 10 ¹⁴	0.00	37,000		[7]
R39	iC ₄ H ₇ + H=aC ₃ H ₄ + + CH ₄		6.31 × 10 ¹³	0.00	0		[7]
R40	iC ₄ H ₇ −1 + H=pC ₃ H ₄ + CH ₄		3.00 × 10 ¹³	0.00	0		b
R41	iC ₄ H ₇ −1=iC ₄ H ₇		5.00 × 10 ¹²	0.00	36,000		[7]
R42	iC ₄ H ₇ =saxC ₄ H ₇						b
		PLOG	9.60 × 10 ¹²	−0.77	47,510	76	
		PLOG	3.00 × 10 ¹¹	0.00	50,000	760	
R43	saxC ₄ H ₇ + H=iC ₄ H ₇ + H						b
		PLOG	6.30 × 10 ¹⁵	−0.26	7600	76	
		PLOG	1.83 × 10 ¹⁸	−0.91	10,079	760	
		PLOG	4.50 × 10 ¹⁸	−1.00	10,756	1520	
		PLOG	5.79 × 10 ¹⁸	−1.01	11,523	3800	
		PLOG	9.30 × 10 ²²	−2.18	14,800	7600	
		PLOG	1.92 × 10 ²⁸	−3.58	21,200	76,000	
R44	C ₃ H ₃ + C ₃ H ₃ =A ₁						[54]
		PLOG	1.07 × 10 ⁴⁵	−9.57	17,015	760	
		PLOG	7.17 × 10 ⁴⁰	−8.24	15,920	7600	
R45	pC ₃ H ₄ + C ₃ H ₃ =A ₁ + H		2.20 × 10 ¹¹	0.00	2000		[55]
R46	aC ₃ H ₄ + C ₃ H ₃ =A ₁ + H		2.20 × 10 ¹¹	0.00	2000		[55]
R47	Fulvene=A ₁						[56]
		PLOG	5.62 × 10 ⁸¹	−19.36	121,500	30	
		PLOG	1.45 × 10 ⁴⁵	−8.90	96,999	760	
		PLOG	2.95 × 10 ³¹	−4.97	88,465	7600	
R48	Fulvene + H=A ₁ + H						[57]
		PLOG	4.94 × 10 ¹⁸	−1.28	5411	1	
		PLOG	2.15 × 10 ²²	−2.28	8429	10	
		PLOG	5.60 × 10 ²⁶	−3.47	12,818	100	
		PLOG	1.66 × 10 ²⁵	−2.99	13,691	760	
		PLOG	5.06 × 10 ²⁵	−3.12	14,226	1000	
		PLOG	2.20 × 10 ²⁷	−3.48	19,199	10,000	
R49	C ₃ H ₃ + C ₃ H ₃ =A ₁ − + H						[56]
		PLOG	5.77 × 10 ³⁷	−7.00	31,506	760	
		PLOG	3.87 × 10 ³³	−5.67	30,411	7600	
R50	C ₃ H ₂ + C ₃ H ₃ =A ₁ −		7.00 × 10 ¹²	0.00	0		[58]
R51	A ₁ CH ₃ (+M)=A ₁ CH ₂ + H(+M)		2.78 × 10 ¹⁵	0.17	91,168		[59]
		LOW /	1.00 × 10 ⁹⁸	−22.86	99,882		
		TROE /	0.0655, 15.11, 1.0 × 10 ¹⁰ , 7.60 × 10 ⁷				
R52	A ₁ CH ₃ + H=A ₁ CH ₂ + H ₂		6.47	3.98	3394		[60]
R53	A ₁ CH ₃ + OH=A ₁ CH ₂ + H ₂ O		1.77 × 10 ⁵	2.39	−602		[61]

(continued on next page)

Table 2 (continued)

	Reactions		A	n	E _a	Pressure [Torr]	Ref.
R54	A ₁ CH ₂ + H = A ₁ + CH ₃						[59]
		PLOG	4.50 × 10 ⁵⁸	−11.90	51,860	30	
		PLOG	2.03 × 10 ⁶⁴	−13.37	59,520	100	
		PLOG	5.83 × 10 ⁶⁷	−14.15	68,330	760	
		PLOG	8.85 × 10 ⁶⁸	−14.23	78,410	7600	
R55	A ₁ + C ₃ H ₃ = C ₃ H ₈		1.50 × 10 ⁷⁵	−17.80	39,600		b
R56	A ₁ + C ₃ H ₃ = C ₃ H ₈ + H		6.26 × 10 ⁹	2.61	56,500		
R57	A ₁ CH ₂ + C ₂ H ₂ = C ₃ H ₈ + H		3.16 × 10 ⁴	2.50	11,061		[64]
R58	A ₁ CH ₂ + C ₃ H ₃ = C ₃ H ₇ CH ₂ + H		5.00 × 10 ¹³	0.00	8000		[27]
R59	C ₉ H ₇ CH ₂ = C ₉ H ₆ CH ₂ + H		7.54 × 10 ³³	−6.23	52,769		[65]
R60	C ₉ H ₇ CH ₂ = A ₂ + H		9.04 × 10 ²⁵	−4.12	43,308		[65]
R61	C ₉ H ₆ CH ₂ + H = A ₂ + H		5.11 × 10 ¹¹	0.63	2422		[65]
R62	C ₇ H ₅ + C ₃ H ₃ = A ₂						[27]
		PLOG	1.82 × 10 ⁷⁴	−18.14	31,896		
		PLOG	3.16 × 10 ⁵⁵	−12.55	22,264		
		PLOG	3.89 × 10 ⁵⁰	−11.01	20,320		
R63	A ₁ + C ₄ H ₄ = A ₂ + H		1.26 × 10 ⁴	2.61	1434		[64]
R64	C ₉ H ₇ + C ₃ H ₃ = A ₂ R ₅ + 2H		8.10 × 10 ⁴²	−9.20	15,153		[27]
R65	A ₂ + C ₂ H ₂ = A ₂ R ₅ + H		8.30 × 10 ³⁰	−5.40	16,900		[46]

^a The rate constants are given as $k = AT^n \exp(-E_a/RT)$ in the units of s, cm³, mol and cal.

^b Proposed in this work, see text.

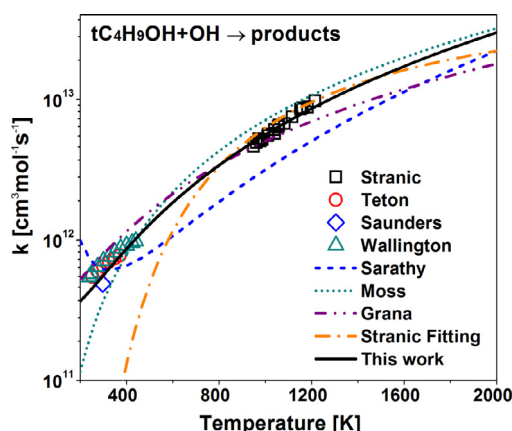


Fig. 2. Total rate constant of H-atom abstraction of tert-butanol by OH radical. The refitted value in the present model is compared to the values in previous models [5,47,48] and experimental data [70–73].

from Stranic et al. [70]. Besides, the H-atom abstraction reactions of tert-butanol by H atom were taken from the model of Grana et al. [48].



tC₄H₈OH is a dominant radical yielded from the decomposition of tert-butanol. Its further β-scission would branch into two channels: via C–O bond scission forming iso-butene (R11), and via C–C bond scission forming iso-propenol (R12). The calculated pressure dependent rate constants by Cai et al. [18] were used for these two reactions. tC₄H₉O radical is less important compared to tC₄H₈OH. Its main consumption pathway is to form acetone (R13). The rate constant proposed by Sarathy et al. [47] was used in the present model.



CH₃COHCH₃ is another important radical yielded from the unimolecular decomposition of tert-butanol. Acetone and iso-propenol isomers can be formed via the β-scission of CH₃COHCH₃. Meanwhile, H atom addition to the unsaturated bonds is considered important in enol chemistry [74]. Therefore, R14 and R15 were included in the present model with their rate constants from the H addition to ethylene. Since iso-propenol can be also formed from tC₄H₈OH radical, the keto–enol tautomerization (R16) [49] might be important in this reaction system. Meanwhile, H-assisted isomerization of iso-propenol (R17) is also considered in the present model according to previous kinetic studies [7,47].

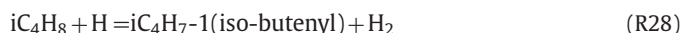
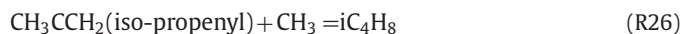


3.2. Sub-mechanism of iso-butene

iso-Butene is the most important intermediate formed during the combustion of tert-butanol. Its sub-mechanism significantly impacts the reactivity and species distribution of this system. Since the production of C₄H₉ isomers from tert-butanol is not efficient [18], the H addition reactions to iso-butene yielding C₄H₉ isomers reveal their importance ((R18) and (R19)). Previous models used either the high-pressure limit (HPL) values of these reactions directly [75], or the estimations from the combination reaction of propene and H atom [50]. In the present model, the rate constants of (R18) and (R19) were proposed in the Troe format, based on the HPL rate constants calculated by Sabbe et al. [76] and recommended by Curran et al. [75], respectively. The Troe coefficients for both reactions were referred to the work of Tsang et al. [50].



The unimolecular decomposition reactions of iso-butene ((R25) and (R26)) were taken from the USC Mech II model [52]. The H-atom abstraction reactions of iso-butene by H, CH₃ and OH radicals ((R27)–(R32)) were mainly taken from the model of Yasunaga et al. [7]. The rate constant of (R29) yielding 2-methyl-allyl radical (iC₄H₇) was referred to the analogous reaction of propene, in which the pre-exponential factor was doubled due to the symmetric methyl structure in iso-butene. The ipso-substitution reaction (R33) between H atom and iso-butene formed propene was considered in the present model according to the USC Mech II model [52]. Meanwhile, similar reaction (R34) by OH radical yielding iC₃H₅OH was also included with an estimated rate constant, although Moss et al. [5] proposed it to form iso-propyl and formaldehyde.



iC₄H₇ and iC₄H₇-1 will form allene and propyne in their further decomposition ((R37)–(R40)). The rate constant of (R37) was from the experimental measurement by Sheen et al. [53]. (R38) and (R39) were taken from the model of Yasunaga et al. [7]. (R40) has the same reaction type as (R39). Its rate constant was estimated as half of (R39)'s in this work.



The isomerization between C₄H₇ radicals, such as iC₄H₇, iC₄H₇-1 and saxC₄H₇ (1-methyl-cyclopropyl) radicals ((R41)–(R43)), was the key step in the formation of 1,3-butadiene, as emphasized by Cai et al. [18]. The isomerization rate constant between iC₄H₇-1 and iC₄H₇ was referred to the recommended value of Yasunaga et al. [7]. The rate constant of the isomerization reactions between iC₄H₇ and saxC₄H₇ ((R42) and (R43)) was estimated in this work. The subsequent decomposition of saxC₄H₇ radical will then form 1,3-butadiene.



3.3. The formation of benzene, phenyl and benzyl radical

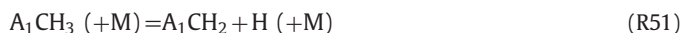
The self-combination reaction of propargyl radical (R44) and its addition reactions to propyne and allene ((R45) and (R46)) are major benzene (A₁) formation. Various rate constants were recommended in previous studies [54,56,77–81]. The rate constant of (R44) in this model was referred to the theoretical investigation of Georgievskii et al. [54], which was modified by Hansen et al. [82] in their study on the premixed flames of C₃H₄ isomers. The addition reactions of propargyl radical to C₃H₄ isomers were referred to D'Anna and Kent [55]. Fulvene can rapidly isomerize to benzene via (R47) and (R48). Their rate constants were taken from the theoretical calculation of Miller and Klippenstein [56] and Jasper and Hansen [57], respectively. Other C₃ + C₃ and C₄ + C₂ formation pathways of benzene and fulvene were also included in the present model.



Phenyl radical (A₁-), an important PAH precursor, could be produced via the H-atom abstraction from benzene and the recombination of small molecules. (R49) is a competitive channel of (R44). Its rate constant was derived from the work of Georgievskii et al. [54]. The reaction of propadienylidene (C₃H₂) and propargyl radicals (R50) was proposed in early kinetic studies [55,58,83] with its rate constant estimated by Laskin et al. [58].



Benzyl radical, another PAH precursor, is mainly formed from toluene and phenyl radical ((R51)–(R54)). The pressure-dependent rate constants of (R51) were taken from the theoretical investigation of Klippenstein et al. [59]. Their results have been experimentally validated in the shock tube pyrolysis of C₆D₅CH₃ by Sivaramakrishnan and Michael [84]. The rate constant of (R52) was taken from the measurement of Oehlschlaeger et al. [60], which is slightly lower than the recommended value in the work of Baulch et al. [85]. The rate constant of (R53) was adopted from the measurements of Seta et al. [61] in the present model. The pressure-dependent rate constant calculated in the work of Klippenstein et al. [59] were used for (R54), which results as another effective benzyl formation channel via its reverse reaction.



3.4. Sub-mechanism of PAH

Indene is mainly formed through (R55)–(R57). The rate of (R55) was estimated according to the values proposed by Ergut et al. [86] and Narayanaswamy et al. [87]. The rate constant of (R56) was

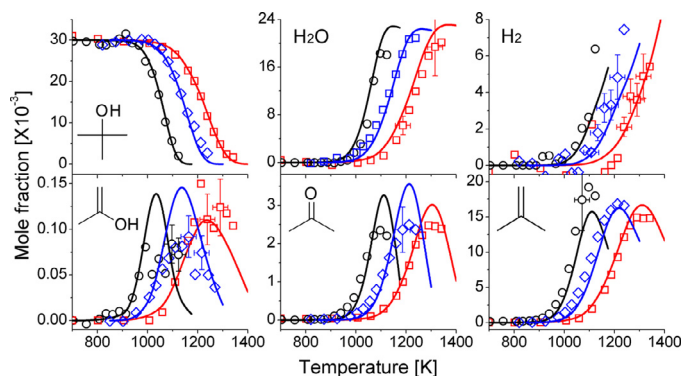


Fig. 3. Model predictions and experimental data of the reactant and major products in the pyrolysis of tert-butanol. The red square, blue diamond and black circle represent the experimental data at 30, 150 and 760 Torr, respectively, and the lines represent the corresponding modeling results. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

referred to the estimation of Matsugi and Miyoshi [62] and Zhang et al. [63] in their modeling studies on the pyrolysis of toluene. The rate constant of the combination of benzyl (A_1CH_2) and acetylene (R57) was referred to the estimation of Blanquart et al. [64], which agreed with the calculation of Vereecken et al. [88,89].



The reaction between of benzyl and propargyl radical is the most efficient formation pathway of naphthalene (A_2) [46,55,64,90,91]. Since it is not easy to restrict the uncertainty of the global reaction rate proposed in previous models, a four-steps kinetic scheme ((R58)–(R61)) [27] was adopted in this model. Methyleneindanyl ($C_9H_7CH_2$) radical is primarily formed via (R58). Its further decomposition could yield naphthalene or methyleneindene ($C_9H_6CH_2$). The latter can isomerize to naphthalene rapidly. The rate constants of (R59)–(R61) were taken from our former study on methane flames [27].



The decomposition of benzyl radical forms fulvenallyl radical (C_7H_5). Its combination with propargyl radicals (R62) results in a considerable naphthalene formation route [27]. The addition of phenyl radical to vinylacetylene (R63) was adopted from the suggestion of Blanquart et al. [64] and Richter and Howard [91]. The self-combination reaction of cyclopentadienyl radicals [55,64,90–92] and HACA mechanism were also included.



(R64) and (R65) are main formation pathways of acenaphthylene (A_2R_5). The recombination of indenyl (C_9H_7) and propargyl radicals (R64) has been proposed in previous PAH models [27,64,90,93]. Its rate constant was taken from the work of Jin et al. [27]. Another important formation pathway of acenaphthylene is (R65). Its rate constant was evaluated by Wang and Frenklach [46,94].



4. Results and discussion

4.1. Flow reactor pyrolysis of tert-butanol

For the pyrolysis of tert-butanol at 30, 150 and 760 Torr, simulations were performed using the plug flow reactor code of the OpenSMOKE program [95]. The experimental and modeling results are presented in Figs. 3 and 4. Good agreement between experimental observations and numerical predictions are achieved for most species, except C_3H_6O isomers. This is because iso-propenol does not have an accurate PICS, which leads to a larger uncertainty in the experimental quantification of itself and its isomer. Detailed kinetic analysis, such as reaction flux analysis and sensitivity analysis, was performed at the fuel conversion ratio around 70% to determine the key reactions and their influences.

Figure 3 presents the decomposition profiles of tert-butanol and its major pyrolysis products, including water, hydrogen, isopropanol, acetone and iso-butene. The unimolecular decomposition

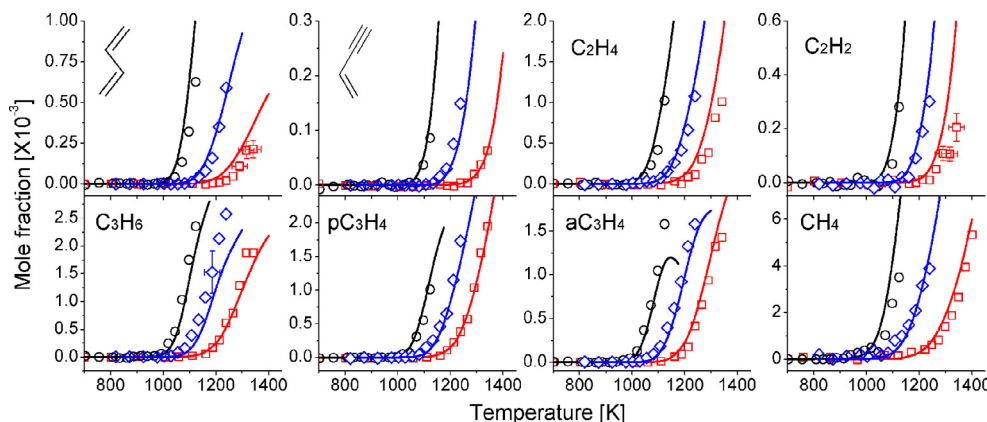


Fig. 4. Model predictions and experimental data of C1–C4 intermediates in the pyrolysis of tert-butanol. The red square, blue diamond and black circle represent the experimental data at 30, 150 and 760 Torr, respectively, and the lines represent the corresponding modeling results. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

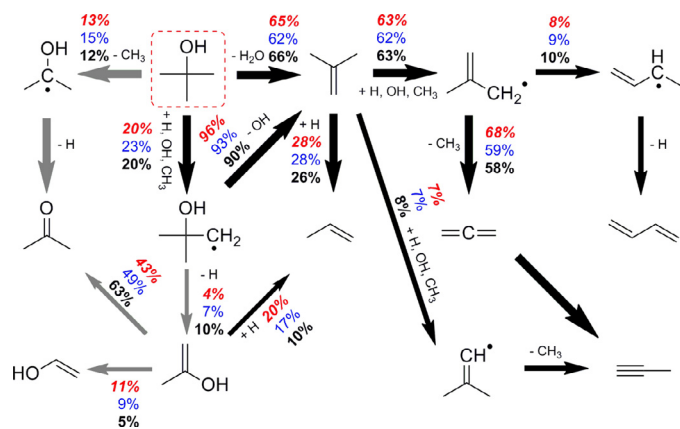


Fig. 5. Decomposition pathways in the pyrolysis of tert-butanol at the fuel conversion of 70%. The percentage labeled beside the arrow is the contribution ratio of specified step. Red italic, blue normal and black bold represent values under 30, 150 and 760 Torr, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

reactions are the initial and dominant reactions. More than 75% of tert-butanol is consumed via the unimolecular decomposition reactions over the whole pressure range of the experiments, as shown in Fig. 5. Among them, water elimination reaction (R1) contributes around 65% to fuel consumption. This high contribution ratio is not seen for any other butanol isomer at similar conditions [30,96,97]. Another significant unimolecular decomposition pathway is the C–C bond scission yielding $\text{CH}_3\text{COHCH}_3$ and CH_3 radicals. The further β -scission of $\text{CH}_3\text{COHCH}_3$ is the main formation pathway of acetone.

H-atom abstraction of tert-butanol forms $\text{C}_4\text{H}_9\text{O}$ radical isomers with $\text{tC}_4\text{H}_8\text{OH}$ dominant. This reaction class totally contributes around 20% in the consumption of tert-butanol. The further decomposition of $\text{tC}_4\text{H}_8\text{OH}$ could lose either an OH group forming iso-butene or a CH_3 group forming iso-propenol. According to the rate of production analysis, the H-atom abstraction of tert-butanol by OH radical contributes around 70% to the formation of $\text{tC}_4\text{H}_8\text{OH}$, while the H-atom abstraction by H atom contributes only 20% (from 24% at 30 Torr to 18% at 760 Torr). In contrast, the contribution of CH_3 radical increases from 2% at 30 Torr to 11% at 760 Torr. The reason for the relatively low contribution of H-atom abstraction by H atom is that the decomposition of tert-butanol cannot efficiently produce large quantity of H atom. The unimolecular C–C bond decomposition of tert-butanol mainly produces CH_3 radical, and the H-atom abstraction reactions of tert-butanol consumes the radical pool, producing CH_3 and OH radicals rather than H atom.

The further decomposition of iso-butene forms small hydrocarbon species. Figure 4 presents the model predictions and the experimental data of C1–C4 stable intermediates. iso-Butene is mainly consumed via the H-atom abstraction reactions forming C_4H_7 radical isomers, the β -scission of which forms either allene or propyne. The ipso-substitution reaction R33 could form propene. The isomerization from iC_4H_7 radical to saxC_4H_7 radical is an important step in the reaction sequence toward the formation of 1,3-butadiene. Zhang et al. [29] proposed this pathway, which critical for explaining the high concentration of 1,3-butadiene observed in the pyrolysis of iso-butene. Later, Cai et al. [96] supported this recommendation in the kinetic investigation on the combustion of iso-butanol. Therefore, this reaction was included in the present model to somehow explain the quantitative measurement of 1,3-butadiene in this work, in spite of the missing support from quantum chemical calculations. Vinylacetylene is formed via 2-step H loss reactions of 1,3-butadiene. Ethylene and acetylene are formed

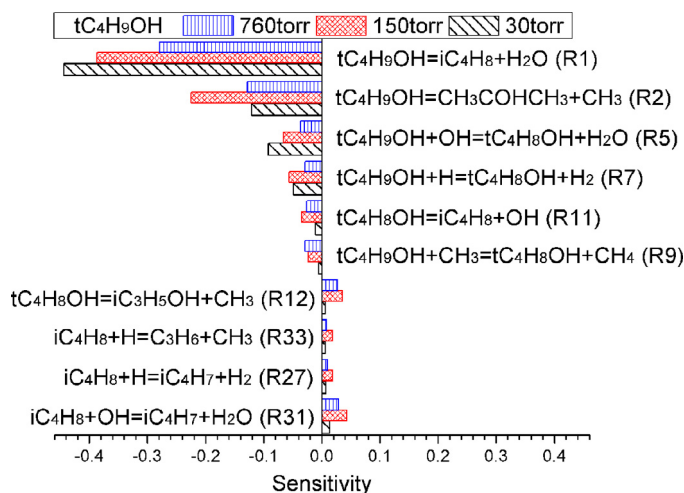


Fig. 6. Sensitivity of tert-butanol at the fuel conversion ratio of 70% in the pyrolysis. Only top 10 reactions are listed here.

via the reaction sequence initiated with the self-combination of methyl radicals.

Figure 6 presents a sensitivity analysis of tert-butanol performed at the same condition of the reaction flux analysis. The consumption reactions of tert-butanol show negative sensitivity coefficients, while the H-atom abstraction reactions of iso-butene reveal positive values. It illustrates a strong competition on free radicals between tert-butanol and iso-butene. There are two reasons for this competition. First, iso-butene is the most abundant hydrocarbon produced from tert-butanol pyrolysis. Reaction flux analysis of tert-butanol reveals an over 80% conversion towards iso-butene. Also the maximum mole fractions of iso-butene reach more than half of the inlet mole fractions of tert-butanol (Fig. 3). Thus, there is a high concentration of iso-butene in the reacting gas mixture. Second, iso-butene has a much weaker C–H bond in CH_3 groups (83.8 kcal/mol [98]) than tert-butanol (103.9 kcal/mol [47]), making its H-atom abstraction favored. Besides, the two decomposition pathways of $\text{tC}_4\text{H}_8\text{OH}$ radical have opposite sensitivity coefficients. Since OH radical is more active in H-atom abstraction reactions than CH_3 radical, the pathway yielding OH radical (R11) has a negative sensitivity coefficient, while the other pathway yielding CH_3 radical (R12) shows a positive value.

4.2. Burner stabilized premixed flame of tert-butanol

Rich burner stabilized tert-butanol flames ($\phi = 1.3$ and 1.8) were investigated in this work, with their inlet conditions presented in Table 1. Simulations were performed using the laminar flame code in the OpenSMOKE program [95]. The temperature profiles used in the simulation and the model prediction of reactants and major products are presented in Fig. 7. C1–C4 hydrocarbon and oxygenated species, as well as some free radicals, methyl, ethyl, propargyl and allyl, were observed in this work. The experimental and predicted mole fractions of these intermediates are compared in Figs. 8 and 9. The model predictions reproduce well the consumption of fuel, oxidizer and final products, as well as most intermediates. But some species, especially free radicals, have large discrepancies between the measured and predicted concentration due to the perturbation of the quartz nozzle and the inaccuracy of PICS.

A rate of production analysis was performed under the rich premixed flame conditions, as shown in Fig. 10. The reaction network in the rich laminar premixed flames (especially in PF2) is similar to that in the low pressure pyrolysis condition. As the equivalence

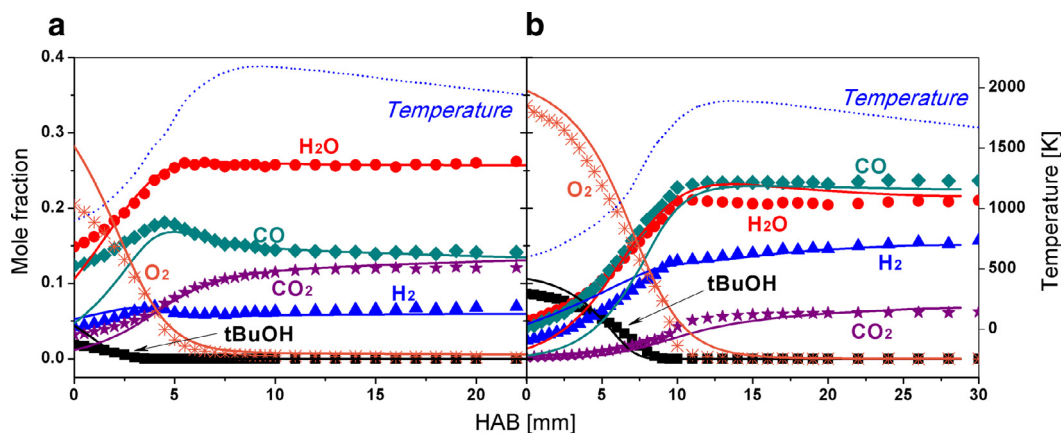


Fig. 7. Model prediction (solid lines) and experimental data (symbols) of major flame species in premixed flames, PF1 (a) (model predictions are shifted for -1 mm for a convenient comparison with experimental data) and PF2 (b).

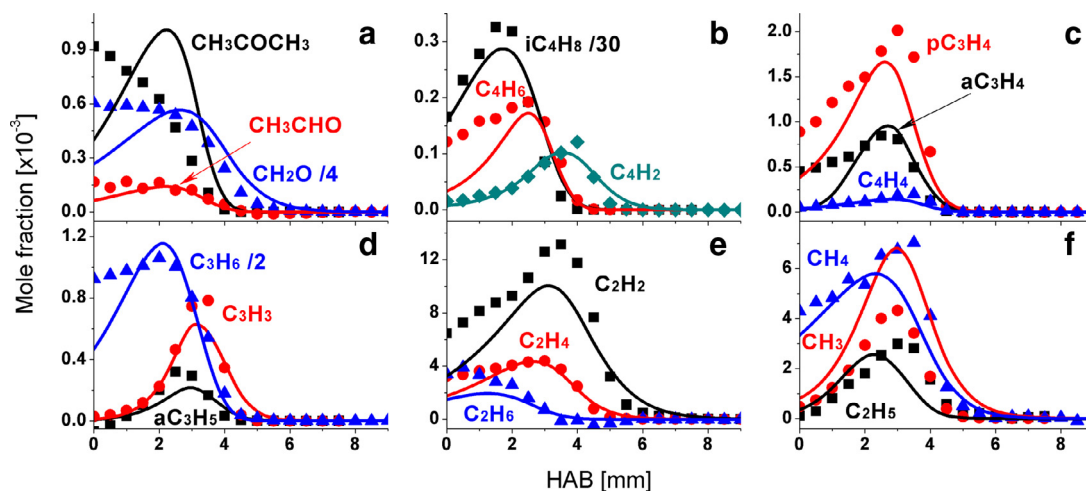


Fig. 8. Model predictions (solid lines, shifted for -1 mm for a convenient comparison with experimental data) and experimental data (symbols) of the flame intermediates in PF1.

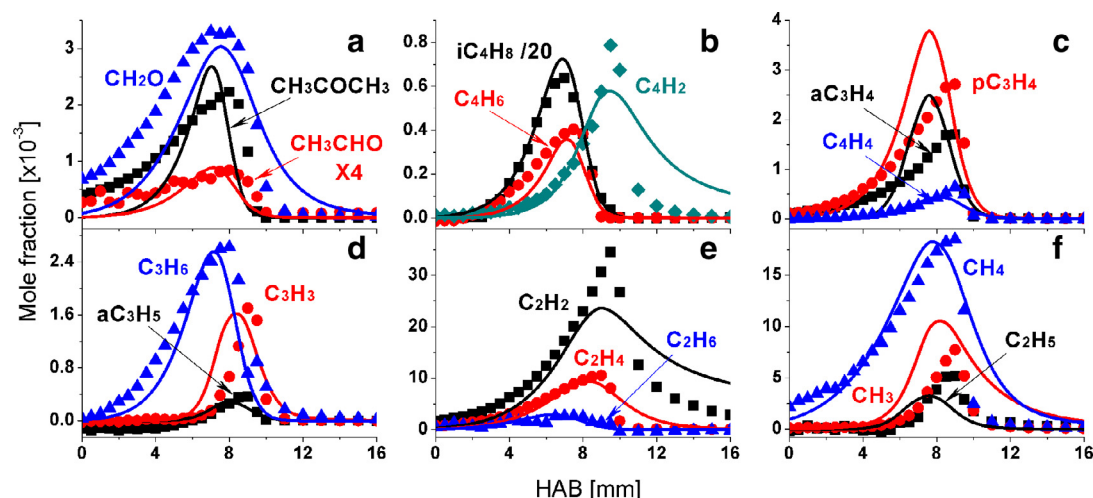


Fig. 9. Model predictions (solid lines) and experimental data (symbols) of the flame intermediates in PF2.

ratio increases, the unimolecular decomposition reactions become more important to the consumption of tert-butanol and the H-atom abstraction reactions have a reverse behavior. Among the H-atom abstraction reactions, those by OH radical play a major role. It consumes around 13% of the fuel in PF2 and more than 30% in PF1, which is around 1.5 times that of H atom. The subsequent

decomposition of iso-butene in the premixed flame does not vary significantly between PF1 and PF2. iC_4H_7 and propene are the dominant products, which will mainly form allene and allyl radical in their further decomposition. H-atom assisted isomerization between iC_4H_7 and $saxC_4H_7$ radical increases its efficiency in premixed flames. But compared to the pyrolysis condition, this

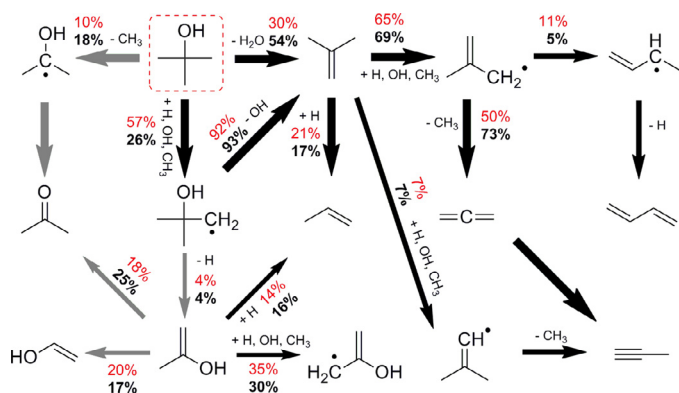


Fig. 10. Comparison of the main reaction pathways in the premixed flames. The conversion of each step is labeled beside arrow. Red normal and black bold represent PF1 and PF2, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

isomerization is not significantly enhanced. Similarly, the isomerization between iso-propenol and acetone is even less efficient in the premixed flames compared to the pyrolysis condition.

4.3. Coflow diffusion flame of methane/tert-butanol

Coflow flames fueled by methane and tert-butanol were also studied in this work, in order to explore the formation mechanism of aromatics in tert-butanol combustion. More than 40 species were quantitatively measured in this experiment. The kinetic modeling of coflow flames was performed with the laminarSMOKE program [99,100], which was introduced in detail elsewhere [27,101]. Both Fickian diffusion and Soret effects [102] were taken into account in the simulation, as well as the radiation heat transfer via CH_4 , H_2O , CO , and CO_2 [103]. A stretched, two-dimensional, rectangular control domain, with a length of 297 mm and a width of 58 mm were adopted in simulation, because of the axial symmetry

of coflow flames. We adopted a finer numerical grid with uniform cell spacing to describe the centerline of the flame near the exit of the burner and the flame region along the radial direction, while used coarser grid points in the flame outer area and air flow. A grid with 9504 cells (144 in flame height and 66 in radial direction) for the simulation was fine enough to balance the accuracy of the calculation and the cost of computational time [27]. The fuel mixture stream was assumed at 493 K with a parabolic velocity profile, while the coflow air was imposed at ambient temperature with a flat velocity profile. Other inlet flow conditions used in the simulation can be found in Table 1. The experimental measurements and numerical simulation results are presented in Figs. 11–13. In the following paragraphs, the major reaction pathways in fuel decomposition and aromatics formation will be analyzed and compared with previous studies of butanol isomers [30,97].

4.3.1. Major flame species

Figure 11 shows the experimental and simulated mole fraction profiles of fuels ($\text{tC}_4\text{H}_9\text{OH}$ and CH_4), O_2 , major products (H_2 , H_2O , CO , CO_2), diluent gas (N_2), and calibration gas (Ar) in the two flames (CF1 and CF2). Figures 12 and 13 present the results of small hydrocarbons and aromatics, respectively. Since the carbon flux in the inlet fuel mixtures was kept identical in CF1 and CF2, the addition of extra 1.95% tert-butanol to the fuel mixture could rarely influence the production of major products. However, for the intermediates that can be formed in the decomposition of tert-butanol or participate in the formation of aromatics, the influence of tert-butanol addition is significant. As discussed in previous work [27,104], the combustion of coflow flame is driven by two major effects, the mass diffusion and the chemical reactions. Therefore, in the flame region with HAB over 20 mm where the gas temperature is beyond 1000 K, reaction kinetics is more influential, while in the bottom region with low temperature, diffusion effect, especially toward radial flame front, strongly impacts the spatial distribution of flame species.

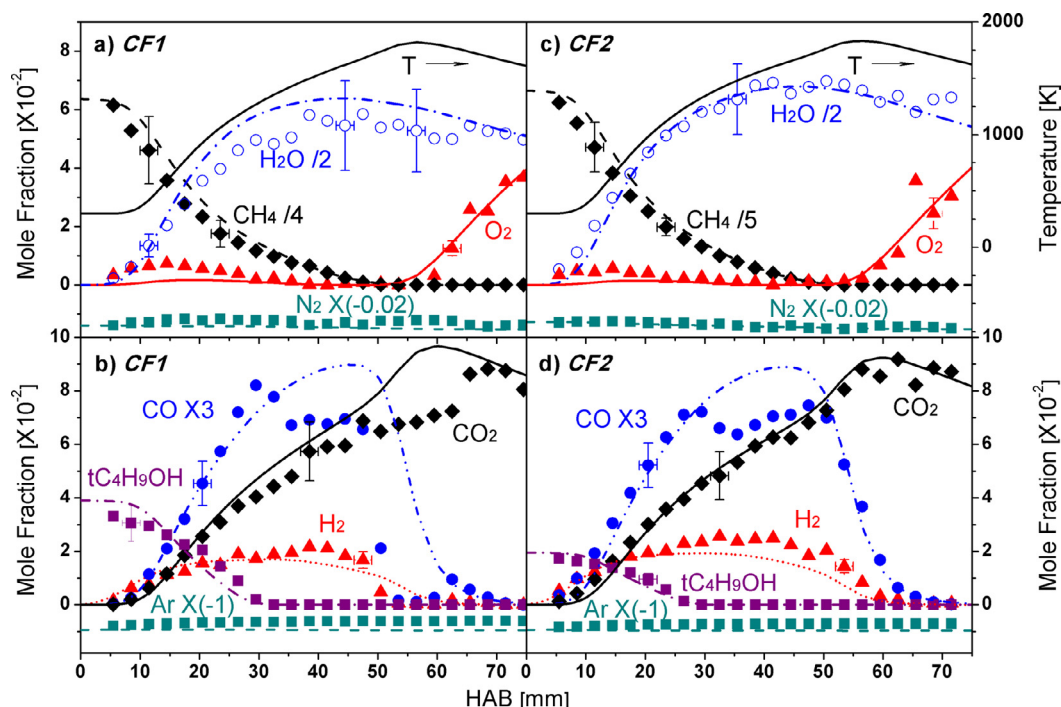


Fig. 11. Model prediction (lines) and experimental data (symbols) of major flame species in coflow diffusion flames, CF1 (a, b) and CF2 (c, d). Temperature profiles in (a) and (c) are given on the right axis.

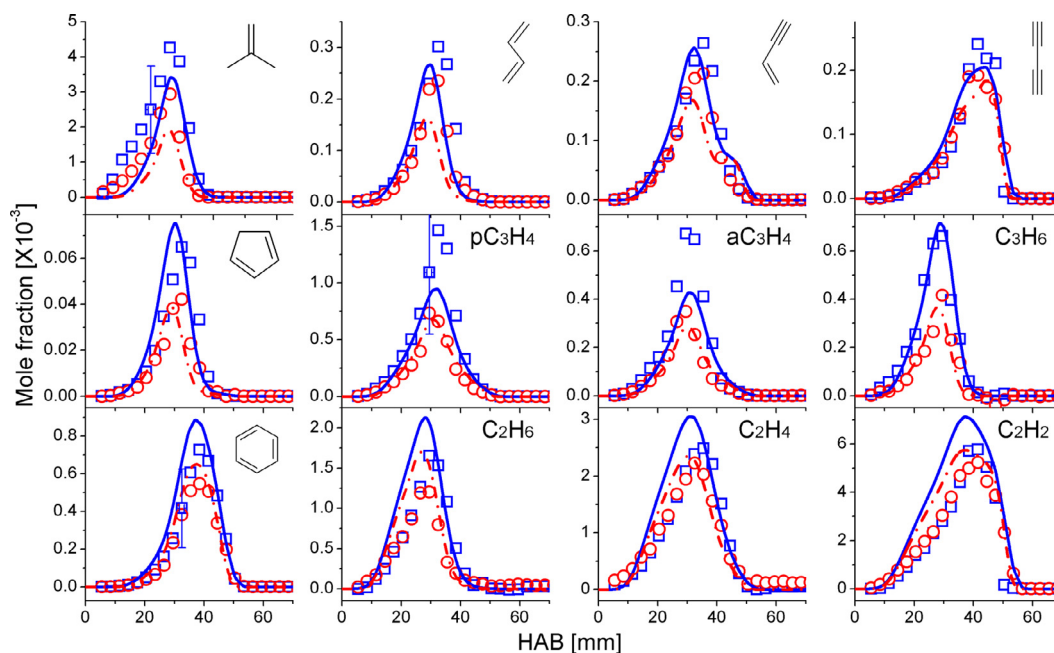


Fig. 12. Model predictions (lines) and experimental data (symbols) of C2–C6 species in coflow diffusion flames. Blue square represents CF1, and red circle represents CF2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

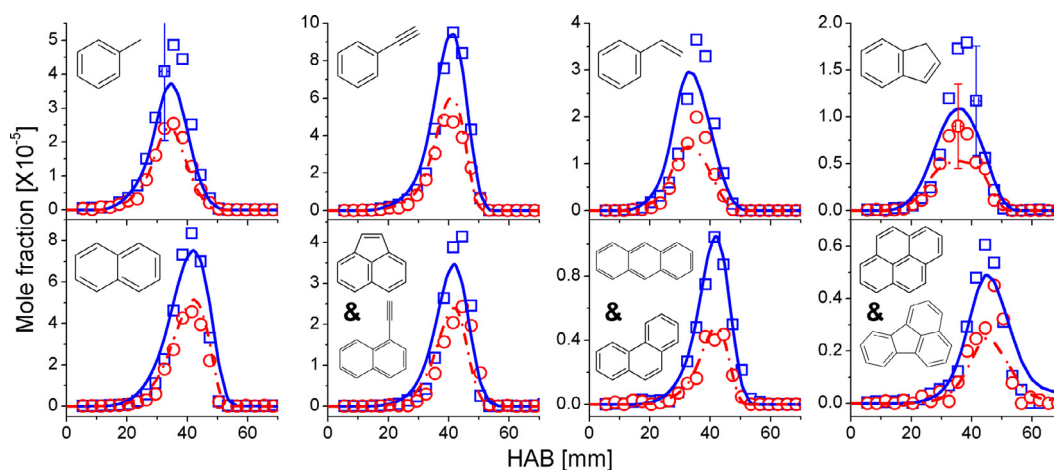


Fig. 13. Model predictions (lines) and experimental data (symbols) of C7–C16 aromatics in coflow diffusion flames. Blue square represents CF1, and red circle represents CF2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3.2. The decomposition of *tert*-butanol

Local ROP analysis along the central axis was performed in CF1 (since the reaction pathways in both flames are very similar), and these values were integrated with HAB until the tip of flame, named as global ROP. Generally, they reveal equivalent conclusions. According to the reaction pathway in Fig. 14, the decomposition procedure of *tert*-butanol in the center axis of coflow flame is very similar to that under pyrolysis condition. Coflow flame is enveloped by its flame front, thus the mass transfer of oxygen molecule or any oxygenated radical cannot easily reach the center of flame. Therefore, it is a proximate pyrolysis environment, and its reaction network is controlled by locally produced active intermediates as emphasized in literatures [27].

Primary decomposition pathways of *tert*-butanol are compared in Fig. 14 under the conditions of coflow flame and pyrolysis. Unimolecular decomposition reactions have even higher contributions in coflow flame. The contribution of water elimination reaction in *tert*-butanol flame (69%) is higher than other butanol isomers (17% in *n*-butanol flame [101] and 45% in *sec*-butanol flame [104]) in the

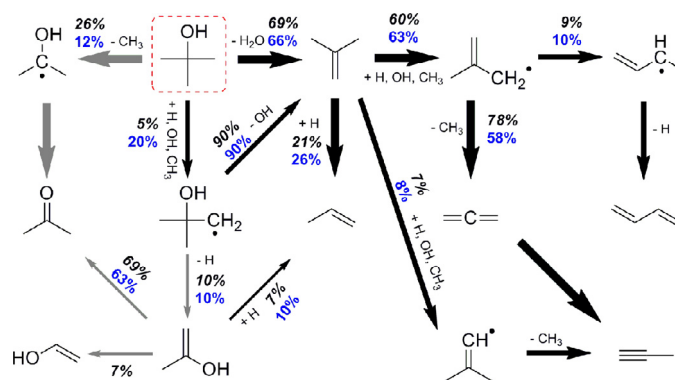


Fig. 14. Comparison of the main reaction pathways in coflow flame (CF1) and the pyrolysis of *tert*-butanol (760 Torr). The conversion of each step is labeled beside arrow, black bold italic, and blue bold represent CF1 and pyrolysis condition, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

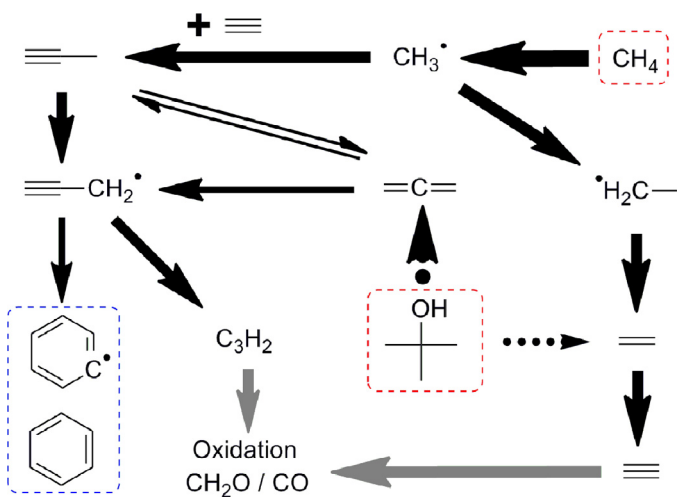


Fig. 15. Major reaction pathways in the formation of the first aromatic ring in CF1. The thickness of the arrow is proportional to the carbon flux of the pathway. The oxidation reactions are highlighted with gray arrows. The dotted arrows denote the simplified decomposition process of tert-butanol.

coflow flames [26]. H-atom abstraction reactions of tert-butanol in coflow flame reveal minor contributions. Just like the competition for H atom between tert-butanol and iso-butene in pyrolysis (Section 4.1), tert-butanol is still the provider of active free radicals in coflow flame, while methane becomes a new radical consumer. CH_3 radical formed via the H-atom abstraction of methane by H atom and OH radical is much less active, which generally reduces the efficiency of H-atom abstraction reactions. As evidence, the H-atom abstraction reactions of iso-butene also reduce their efficiencies due to the same effect.

iso-Butene is the dominant decomposition product of tert-butanol, and acetone is a subordinate one. Since the investigation of coflow flame is mainly focused on the formation of PAHs, the pathways for oxygenated species will not be discussed in detail in the following paragraphs. Their model predictions are provided in the Supplementary material, including acetone, iso-propenol, acetaldehyde, and formaldehyde. The further decomposition pathway of iso-butene has been introduced in pyrolysis (Section 4.1), which forms allene, propyne, propene and 1,3-butadiene, etc. The H-atom abstraction of allene and propyne could form propargyl radical. However, this pathway only plays an additional role, as shown in Fig. 15. The main formation pathway of propargyl radical is the decomposition of propyne, which is formed through the combination of acetylene and methyl radical [101,104].

4.3.3. The formation of aromatics

According to the reaction flux analysis, benzene is formed mainly via the self-combination of propargyl radical. Its H-atom abstraction reaction could form phenyl radical, which greatly promote the growth of aromatics. Monocyclic aromatics, such as toluene, styrene, and phenylacetylene, can be formed via the addition of methyl, ethylene and acetylene to phenyl radical. According to the reaction flux analysis in Fig. 16, 53% of toluene is formed via the reaction of benzene and methyl radical, while mainly converts to benzyl radical. The main consumption channels of phenyl radical are the addition of methyl (R54) and propargyl radicals forming benzyl radical and indene, respectively. The decomposition of toluene and (R54) have almost identical contributions in the formation of benzyl radical, while the combination of propargyl radical and vinylacetylene plays a minor role. For styrene, the combination of cyclopentadienyl and propargyl radicals ($\text{C}_5\text{H}_5 + \text{C}_3\text{H}_3 \rightarrow \text{C}_6\text{H}_5\text{C}_2\text{H}_3 + 2\text{H}$, (R66)) is also efficient beside the addition of ethylene to phenyl radical. This pathway is not

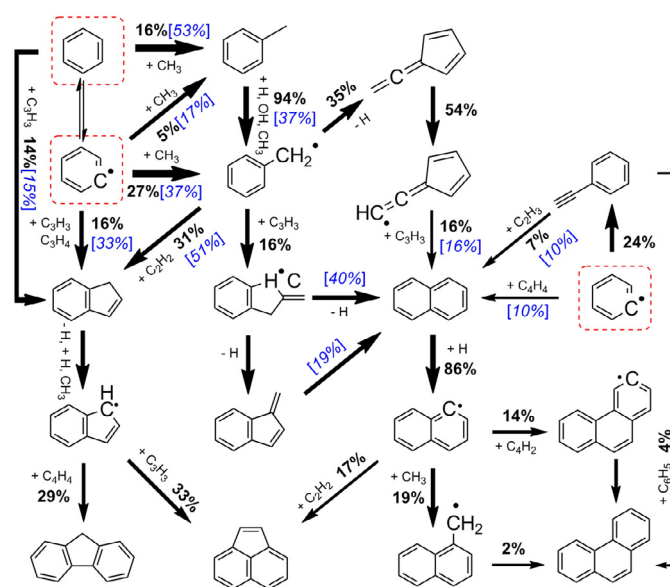


Fig. 16. Major reaction pathways in the formation of large aromatic species in CF1. The thickness of the arrow is proportional to the carbon flux of the pathway. The number beside the arrow is the conversion ratio of this reaction step, black normal and blue in bracket indicate the contribution in decomposition and formation of the related species. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

important in the methane coflow flame [27]. However, its importance has increased with the addition of tert-butanol. The reaction flux analysis of styrene shows that R66 has a contribution of 46% to styrene formation. Cyclopentadienyl radical is derived from 1,3-cyclopentadiene, which is formed via the combination of acetylene and allyl radical. The doping of tert-butanol increases the concentration of allyl radical through the following reaction sequence, $\text{tC}_4\text{H}_9\text{OH} \rightarrow \text{C}_4\text{H}_8 \rightarrow \text{C}_3\text{H}_6 \rightarrow \text{aC}_3\text{H}_5$.

Polycyclic aromatic hydrocarbons, such as indene, naphthalene, acenaphthalene/ethylnaphthalene, phenanthrene/anthracene, and pyrene/fluoranthene, were observed in this work. As shown in Fig. 13, the present model can generally reproduce the experimental results of these PAHs well. Indene is formed via the reactions of benzene, phenyl and benzyl radical with small intermediates. The reactions of phenyl radical with C3 intermediates (propyne, allene and propargyl radical) and the combination of benzyl radical and acetylene contribute 33% and 51% in CF1, respectively. The combination of propargyl and benzene has a relatively high contribution, because of the insufficient H-atom abstraction of benzene and the accumulation of propargyl radical in the center of coflow flame [27]. Benzyl, phenyl, cyclopentadienyl radical, and phenylacetylene are precursors of naphthalene. Among them, the combination of benzyl and propargyl radical is the dominant pathway with a contribution of 60%. Benzyl radical can also undergo another pathway to form naphthalene via an intermediate radical of C_7H_5 , as shown in Fig. 16. The combination of phenyl and vinylacetylene and the HACA pathway [46,94] are minor pathways in this flame condition, while the contribution from the self-combination of cyclopentadienyl radical is comparatively negligible. Indenyl and naphthyl radicals are the precursors of tricyclic and tetracyclic PAHs. The combination of indenyl and propargyl radicals is the dominant formation pathway of acenaphthalene, as emphasized in [105]. And phenanthrene is mainly formed through the combination of indenyl and cyclopentadienyl radicals, which was also recommended in [64,106,107]. The dominant precursors of fluoranthene and pyrene are naphthyl and acenaphthalene in the present model.

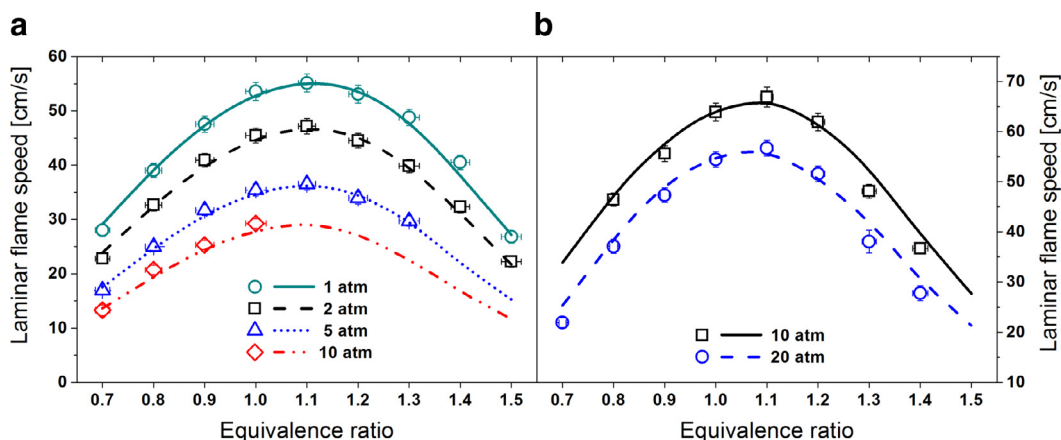


Fig. 17. Model predictions and experimental measurements of laminar flame speeds of tert-butanol at $T_u = 423$ K. (a) Air was used as the oxidizer; (b) 14%O₂/86%He was used as the oxidizer and inert gas.

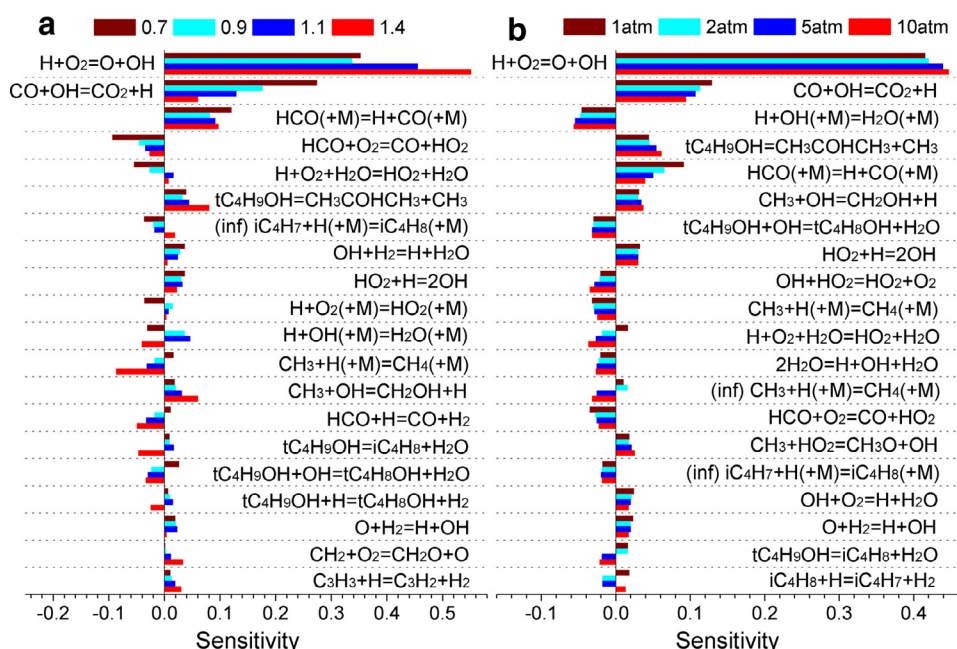


Fig. 18. Sensitive reactions for laminar flame propagation for tert-butanol: (a) under the same pressure of 1.0 atm; and (b) in the same equivalence ratio of 1.1.

In summary, the sooting tendency of coflow flame is generally determined by the efficiency of the formation of propargyl radical through the decomposition of the doped butanol isomer [101,104]. PAH precursors, like benzene, phenyl and benzyl, are all formed based on the self-combination of propargyl radical. Furthermore, the addition reactions of propargyl radical to these precursors are also the key formation pathways for heavier aromatic species. tert-Butanol has a high efficiency pathway for propargyl radical: $tC_4H_9OH \rightarrow iC_4H_8 \rightarrow iC_4H_7 \rightarrow aC_3H_4 \rightarrow C_3H_3$; therefore, the amount of PAHs increase significantly with the increment of its doping ratio.

4.4. Flame propagation of tert-butanol

Laminar flame speed of tert-butanol at various pressures (1–20 atm) was measured in a single cylinder combustion vessel. The experimental conditions are presented in Table 1. Four groups of measurements used air as the oxidizer, while the other two groups at higher pressures used oxygen and helium as oxidizer and inert gas. Kinetic modeling under these conditions was performed using the laminar flame code of the OpenSMOKE program [95] in

this work. The model predictions are compared with experimental data in Fig. 17. Since the sub-mechanism of iso-butene plays a crucial role in the tert-butanol model, the laminar flame speeds of iso-butene reported by Zhao et al. [108] were also used to validate the present model, as shown in Fig. S3 in the Supplementary material.

As shown in Fig. 17, the dependencies of laminar flame speed to temperature and pressure are well captured by the present model. Sensitive reactions to laminar flame speed of tert-butanol at different equivalence ratios ($P = 1.0$ atm) and pressures ($\phi = 1.1$) are listed in Fig. 18. The oxidation reactions of H atom and HCO radical possess high sensitivities at lean conditions, while the reactions of CH₃ and tC_4H_9OH are important at rich conditions. The unimolecular C–C bond dissociation of tert-butanol (R2) is an important initial fuel decomposition reaction that produces free radicals. Its sensitivity increases with the equivalence ratio. In contrast, the unimolecular water elimination reaction of tert-butanol (R1) has smaller sensitivity coefficient than (R2). It is even negative under extremely rich condition, because it produces stable iso-butene and water instead of free radicals. The kinetic modeling performance shown in Fig. 17 reflects the reasonable estimations

of their rate constants at high pressure (Section 3.1). The chain propagation reaction that converts a CH_3 radical into an H atom ($\text{CH}_3 + \text{OH} = \text{CH}_2\text{OH} + \text{H}$, (R67)) is very sensitive in the rich flame condition, whilst the termination of CH_3 radical is also sensitive but with a negative coefficient.

Sensitive reactions also significantly changes with pressure, particularly the unimolecular decomposition reactions of tert-butanol. Usually the pressure dependence of elementary reactions involving small species are more obvious, such as HCO, H, OH and CH_3 , etc. [109]. In tert-butanol flame, (R2) becomes more sensitive as the pressure increases, while (R1) has a negative coefficient at high pressure. The decomposition of tert-butanol via (R2) and the further β -scission of $\text{CH}_3\text{COHCH}_3$ could yield H atom and CH_3 radical, thus is favored. The H-atom abstraction reaction of tert-butanol by OH radical (R5) has relatively strong negative sensitivities, but does not vary too much at different pressures. It consumes an OH radical, but produces a CH_3 or OH radical, which generally reduces the reactivity of the system.

5. External model validations

The present model was also validated using the experimental data in literatures [2,5,7,17,19–22]. The comparison of modeling and experimental results is provided in the Supplementary material. Experimental data of homogeneous ideal reactors are valuable for the validation of primary decomposition reactions of tert-butanol. It helps to constrain the rate constants of the unimolecular decomposition, H-atom abstraction and β -scission reactions. Modeling result of the shock tube pyrolysis of tert-butanol in the pressure range of 0.7–1.2 atm [21] is presented in Fig. S4. And the simulation of the flow reactor pyrolysis and oxidation of tert-butanol studied by Norton et al. [22] and Lefkowitz et al. [19] at atmospheric pressure can be found in Fig. S5. The experimental data of laminar flames were also used to validate the present model. It focuses on the reaction network in the formation of the combustion intermediates. The modeling results of the low-pressure rich premixed flame studied by Oßwald et al. [17] and the atmospheric pressure counterflow diffusion flame studied by Lefkowitz et al. [19] are presented in Figs. S6 and S7. Besides, the laminar flame speeds measured by Wu et al. [2], and the ignition delay times measured by Moss et al. [5] and Stranic et al. [20] under various pressure and equivalence ratio conditions were also simulated. The modeling results of their data are shown in Figs. S8–S10. Macro combustion parameters generally reflect the reactivity of tert-butanol in different combustion conditions, which are also well reproduced by the present model.

6. Conclusions

In this work, a series of experiments were performed to investigate the combustion chemistry of tert-butanol, including flow reactor pyrolysis in 30–760 Torr, burner stabilized premixed flames at 30 Torr, coflow diffusion flames at atmospheric pressure, and flame speed measurement in the pressure range of 1–20 atm. A kinetic model was developed to reproduce the high temperature combustion of tert-butanol over a wide pressure range. The fuel decomposition process as well as the PAH formation mechanism was analyzed in detailed based on the kinetic modeling results.

Under the pyrolysis and flame conditions, unimolecular decomposition reaction is the dominant reaction class among the fuel consumption pathways, in which four center ring water elimination reaction has extremely high contribution. iso-Butene and acetone are the main primary products. The further decomposition of iso-butene yields a series of small hydrocarbons. Therefore, the main reaction routine for the decomposition of tert-butanol is

$\text{tC}_4\text{H}_9\text{OH} \rightarrow \text{iC}_4\text{H}_8 \rightarrow \text{iC}_4\text{H}_7 \rightarrow \text{aC}_3\text{H}_4$. In the coflow flames, this reaction sequence is an efficient tert-butanol decomposition pathway for the production of propargyl radical, which in turn leads to the formation of large amount of PAHs. Propargyl is a major precursor of aromatic species, phenyl and benzyl radicals are main PAH precursors, or even indenyl and naphthyl radicals for larger PAHs. The self-combination of propargyl and its combination with other resonantly stabilized radicals are the key steps for aromatics formation in the combustion of tert-butanol. Finally, according to the study of laminar flamed speed, the most sensitive reaction that involves tert-butanol is the unimolecular decomposition reaction forming $\text{CH}_3\text{COHCH}_3$ and CH_3 radicals (R2).

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2016.03.026.

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